

Game Theory – 2

Strategy stealing, pairing and invariants; games with a value

Homework

14 сентября 2026 года

Theoretical minimum. Outside the impartial finite games of Game Theory – 1 four arguments do the work. Strategy stealing: in a symmetric game with no draws in which an extra move never hurts, the first player wins, because otherwise he could steal the second player’s strategy — the proof gives no winning move and is the standard example of non-constructive existence. Pairing and symmetry: if the positions split into pairs so that the second player can always answer inside the pair just used, he does not lose. Invariants and monovariants: a preserved quantity separates reachable positions, a strictly decreasing bounded quantity proves that the game ends, which is what backward induction needs. Finally, for a finite zero-sum game with payoff matrix A the minimax theorem says $\max_x \min_y x^\top A y = \min_y \max_x x^\top A y$ over mixed strategies, the common number being the value of the game; it follows from the separation theorem for convex sets. For infinite games determinacy must be earned: open games are determined (Gale–Stewart), and with the axiom of choice one builds a game that is not.

1. Alice and Bob take turns placing tiles covering two adjacent squares on a row of 2022 squares, Alice first, and a tile may not overlap an earlier one; the game ends when no tile can be placed. Alice wants to maximise the number of uncovered squares, Bob to minimise it. What is the greatest number of uncovered squares Alice can guarantee? (Putnam 2022)
2. Alice and Bob take turns choosing integers from 1 to n , all distinct, Alice first, until all n have been chosen. Before the game Bob announces a goal, “odd” or “even”. Bob wins if the number of indices k such that

k was chosen on the k -th turn has the announced parity. Who wins, and for which n ? (Putnam 2023)

3. Two players take turns choosing an element of the group of invertible $n \times n$ matrices over $\mathbb{Z}/p\mathbb{Z}$, never repeating an element already chosen and always choosing an element that commutes with all previously chosen elements. The player with no legal move loses. Determine the winner in terms of n and p . (Putnam 2014)
4. Let $n > 1$ and let S_n be the group of permutations of $\{1, \dots, n\}$. Two players take turns selecting elements of S_n , never repeating one already selected. The game ends as soon as the selected elements generate the whole of S_n , and the player who made the last move loses. Which player has a winning strategy? (IMC)
5. Let $n, k \geq 1$. Alice places each of k coins into one of n jars, choosing for each coin whether it lies heads up or tails up. Bob may then repeatedly take any coin from any jar and move it to another jar, but each time Alice chooses which way up it lands. Bob wins if he reaches a state in which inside every jar all coins show the same face. Find all n, k for which Bob can win. (IMC 2025, shortlist)
6. Let $(G, *)$ be a group with $n > 1$ elements and let $g \in G$ be different from the identity. Ana and Bob alternately select elements of G not selected before, Ana first, until all elements are gone or some player has selected both a and $a * g$ for some a — in which case that player loses. (a) If n is odd, show that one of the players has a winning strategy regardless of g , and say which. (b) If n is even, show that there is g for which neither player has a winning strategy. (CIIM 2019)
7. Two players alternately place dominoes on an $m \times n$ board, each domino covering two adjacent cells and not overlapping an earlier one; the player unable to move loses. Prove that if m and n are both even the second player wins, and that if exactly one of them is odd the first player wins. Then show that the same pairing idea fails when both are odd, and explain what the obstruction is.

8. Three copies of the number 2024 are written on a blackboard. The Mad Hatter and Alice move alternately, the Hatter first: he picks one of the numbers and increases it by 1; Alice then picks one of the numbers and decreases it by 1, after which she replaces the two numbers not touched by the Hatter on his previous move by two copies of their geometric mean. Alice wins if at the start of one of her turns some number is less than 1. Can the Hatter prevent this? (ICMC)
9. A computer picks a number uniformly at random from 1 to N . Two players alternately name numbers from 1 to N not named before; after each attempt the computer says whether the hidden number is equal to, greater than, or smaller than the number named. The winner is the player who names the hidden number. Both maximise their probability of winning. Find the probability that the first player wins. (Mexmat 2014)
10. Prove the Gale–Stewart theorem: if the winning set of the first player is open, then the game of infinite length is determined. Deduce that the game in which the players alternately choose digits and the first player wins if the resulting number lies in a fixed open subset of $[0, 1]$ is determined, and explain why the same argument fails for an arbitrary set, quoting the construction by transfinite recursion of a non-determined game (Topology of the real line – 3).